

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[4\sqrt{2}(\cos 165^\circ + i \sin 165^\circ) \right] \left[3\sqrt{2}(\cos 240^\circ + i \sin 240^\circ) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

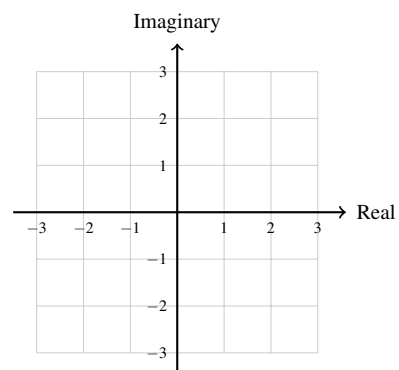
2. $\frac{-3\sqrt{6} + 3i\sqrt{2}}{\sqrt{3} - 3i}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $(\sqrt{6} - i\sqrt{2})^5$

Find the fourth roots of the number. Leave answers in trig form. Graph each root as a vector in the complex plane. (3 points)

4. $-8 + 8i\sqrt{3}$



Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

5. $x^3 + 27i = 0$

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!*Every line shown. Point splits are an estimate of partial credit (halves are possible).*

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[4\sqrt{2}(\cos 165^\circ + i \sin 165^\circ)\right] \left[3\sqrt{2}(\cos 240^\circ + i \sin 240^\circ)\right]$

1. Multiply the r 's, add the θ 's.

$$\begin{aligned} r &= (4\sqrt{2})(3\sqrt{2}) = 12 \cdot 2 = 24 & \theta &= 165^\circ + 240^\circ = 405^\circ \\ 405^\circ - 360^\circ &= 45^\circ & & \text{(keep the angle between } 0^\circ \text{ and } 360^\circ) \\ 24(\cos 45^\circ + i \sin 45^\circ) &= 24\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) \\ &= \boxed{12\sqrt{2} + 12\sqrt{2}i} \end{aligned}$$

*Trap: $\sqrt{2} \cdot \sqrt{2} = 2$, so $r = 12 \cdot 2 = 24$. Leaving $12\sqrt{4}$, or writing $12\sqrt{2}$, costs the point.**1 pt r, 1 pt reduced angle, 1 pt exact rectangular.*

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

Problem: $\frac{-3\sqrt{6} + 3i\sqrt{2}}{\sqrt{3} - 3i}$

2. Convert the top and the bottom separately (sketch each for the quadrant).

Top: $r = \sqrt{(-3\sqrt{6})^2 + (3\sqrt{2})^2} = \sqrt{54 + 18} = \sqrt{72} = 6\sqrt{2}$ $\tan \theta = \frac{3\sqrt{2}}{-3\sqrt{6}} = -\frac{1}{\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QII} \Rightarrow 150^\circ$

Bottom: $r = \sqrt{(\sqrt{3})^2 + (-3)^2} = \sqrt{3 + 9} = \sqrt{12} = 2\sqrt{3}$ $\tan \theta = \frac{-3}{\sqrt{3}} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ$

$$\begin{aligned} \frac{6\sqrt{2} \text{ cis } 150^\circ}{2\sqrt{3} \text{ cis } 300^\circ} &= \frac{3\sqrt{2}}{\sqrt{3}} \text{ cis}(150^\circ - 300^\circ) = \frac{3\sqrt{6}}{3} \text{ cis}(-150^\circ) = \sqrt{6} \text{ cis}(-150^\circ + 360^\circ) = \sqrt{6} \text{ cis } 210^\circ \\ &= \sqrt{6} \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = -\frac{\sqrt{18}}{2} - \frac{\sqrt{6}}{2}i = \boxed{-\frac{3\sqrt{2}}{2} - \frac{\sqrt{6}}{2}i} \end{aligned}$$

Trap: $(-3\sqrt{6})^2 = 9 \cdot 6 = 54$ (square the 3 too). Negative angle: add 360° . Rationalize $\frac{3\sqrt{2}}{\sqrt{3}}$ and simplify $\sqrt{18} = 3\sqrt{2}$.

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $(\sqrt{6} - i\sqrt{2})^5$

3. Convert, then De Moivre: $[r \operatorname{cis} \theta]^n = r^n \operatorname{cis}(n\theta)$.

$$r = \sqrt{(\sqrt{6})^2 + (-\sqrt{2})^2} = \sqrt{6+2} = \sqrt{8} = 2\sqrt{2}$$

$$\tan \theta = \frac{-\sqrt{2}}{\sqrt{6}} = -\frac{1}{\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QIV} \Rightarrow 330^\circ$$

$$(2\sqrt{2} \operatorname{cis} 330^\circ)^5 = (2\sqrt{2})^5 \operatorname{cis}(5 \cdot 330^\circ) = 2^5 (\sqrt{2})^5 \operatorname{cis} 1650^\circ$$

$$= 32 \cdot 4\sqrt{2} \operatorname{cis} 1650^\circ = 128\sqrt{2} \operatorname{cis} 1650^\circ$$

$$1650^\circ - 4(360^\circ) = 1650^\circ - 1440^\circ = 210^\circ$$

$$128\sqrt{2}(\cos 210^\circ + i \sin 210^\circ) = 128\sqrt{2} \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \boxed{-64\sqrt{6} - 64\sqrt{2}i}$$

Trap: $(\sqrt{2})^5 = \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} = 4\sqrt{2}$, not $5\sqrt{2}$ or $2\sqrt{5}$.

Find the fourth roots of the number. Leave answers in trig form. Graph each root as a vector in the complex plane. (3 points)

Problem: $-8 + 8i\sqrt{3}$

4. Convert, then the n th root theorem: $\sqrt[n]{r} \operatorname{cis} \left(\frac{\theta}{n} + \frac{360^\circ k}{n} \right), k = 0, 1, \dots, n-1$.

$$r = \sqrt{(-8)^2 + (8\sqrt{3})^2} = \sqrt{64 + 192} = \sqrt{256} = 16$$

$$\tan \theta = \frac{8\sqrt{3}}{-8} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QII} \Rightarrow 120^\circ$$

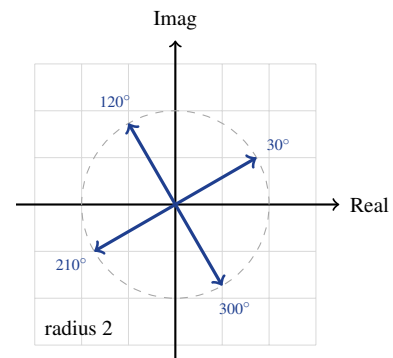
$$\sqrt[4]{16} = 2 \quad \text{first angle } \frac{120^\circ}{4} = 30^\circ$$

$$\text{spacing } \frac{360^\circ}{4} = 90^\circ$$

$$\boxed{2 \operatorname{cis} 30^\circ, 2 \operatorname{cis} 120^\circ, 2 \operatorname{cis} 210^\circ, 2 \operatorname{cis} 300^\circ}$$

Stop at four roots: the next one, 390° , is 30° again.

Trap: $\sqrt{256} = 16$, then $\sqrt[4]{16} = 2$ (not 4).



four equal vectors, 90° apart

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

Problem: $x^3 + 27i = 0$

5. Isolate x^3 first, then take cube roots of the constant.

$$x^3 + 27i = 0 \Rightarrow x^3 = -27i = 0 - 27i \quad (\text{negative imaginary axis})$$

$$r = 27, \quad \theta = 270^\circ \quad x = (27 \operatorname{cis} 270^\circ)^{1/3}$$
$$\sqrt[3]{27} = 3 \quad \frac{270^\circ}{3} = 90^\circ \quad \text{spacing } \frac{360^\circ}{3} = 120^\circ$$
$$x = 3 \operatorname{cis} 90^\circ, 3 \operatorname{cis} 210^\circ, 3 \operatorname{cis} 330^\circ$$

$$3 \operatorname{cis} 90^\circ = 3(0 + i) = 3i$$

$$3 \operatorname{cis} 210^\circ = 3\left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = -\frac{3\sqrt{3}}{2} - \frac{3}{2}i$$

$$3 \operatorname{cis} 330^\circ = 3\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = \frac{3\sqrt{3}}{2} - \frac{3}{2}i$$

$$x = 3i, -\frac{3\sqrt{3}}{2} - \frac{3}{2}i, \frac{3\sqrt{3}}{2} - \frac{3}{2}i$$

Check one: $(3i)^3 = 27i^3 = 27(-i) = -27i \checkmark$

Trap: moving $+27i$ across makes it $-27i$, which is 270° , not 90° . Using 90° gives the wrong three roots.

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Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[2 \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[5 \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

Find the quotient. Write it in rectangular form, using exact values. (3 points)

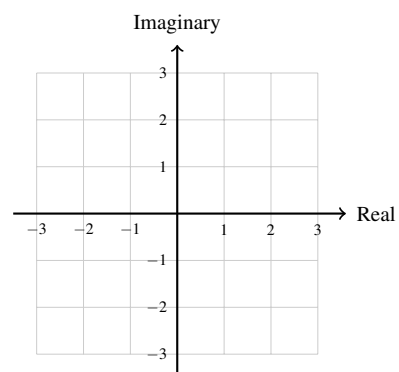
2. $\frac{\sqrt{7} + i\sqrt{21}}{-3\sqrt{21} - 3i\sqrt{7}}$

Find the power. Write the answer in rectangular form. (3 points)

3. $(-2 + 3i)^4$

Find the fifth roots of the number. Leave answers in trig form with θ in radians. Graph each root as a vector in the complex plane. (3 points)

4. $16 - 16i\sqrt{3}$



One of the fourth roots of a complex number w is $-\sqrt{2} + i\sqrt{2}$. (3 points)

5. a) Determine the rectangular form of its other three fourth roots.

b) Find w in rectangular form.

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Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[2 \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[5 \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

1. Multiply the r 's, add the θ 's (common denominator first).

$$r = 2 \cdot 5 = 10 \quad \theta = \frac{5\pi}{4} + \frac{11\pi}{12} = \frac{15\pi}{12} + \frac{11\pi}{12} = \frac{26\pi}{12} = \frac{13\pi}{6}$$

$$\frac{13\pi}{6} - 2\pi = \frac{13\pi}{6} - \frac{12\pi}{6} = \frac{\pi}{6} \quad (\text{keep between } 0 \text{ and } 2\pi)$$

$$10 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 10 \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = \boxed{5\sqrt{3} + 5i}$$

Trap: given in radians, stay in radians. $2\pi = \frac{12\pi}{6}$, so subtract $\frac{12\pi}{6}$, not $\frac{2\pi}{6}$.

Find the quotient. Write it in rectangular form, using exact values. (3 points)

Problem: $\frac{\sqrt{7} + i\sqrt{21}}{-3\sqrt{21} - 3i\sqrt{7}}$

2. Convert top and bottom, divide the r 's, subtract the θ 's.

Top: $r = \sqrt{(\sqrt{7})^2 + (\sqrt{21})^2} = \sqrt{7 + 21} = \sqrt{28} = 2\sqrt{7}$ $\tan \theta = \frac{\sqrt{21}}{\sqrt{7}} = \sqrt{3} \Rightarrow \text{QI}, 60^\circ$

Bottom: $r = \sqrt{(-3\sqrt{21})^2 + (-3\sqrt{7})^2} = \sqrt{189 + 63} = \sqrt{252} = 6\sqrt{7}$ $\tan \theta = \frac{-3\sqrt{7}}{-3\sqrt{21}} = \frac{1}{\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QIII} \Rightarrow 210^\circ$

$$\frac{2\sqrt{7} \operatorname{cis} 60^\circ}{6\sqrt{7} \operatorname{cis} 210^\circ} = \frac{1}{3} \operatorname{cis}(60^\circ - 210^\circ) = \frac{1}{3} \operatorname{cis}(-150^\circ) = \frac{1}{3} \operatorname{cis} 210^\circ$$

$$= \frac{1}{3} \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \boxed{-\frac{\sqrt{3}}{6} - \frac{1}{6}i}$$

Trap: both parts negative means QIII. $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ says 30° ; the sketch says 210° . $\sqrt{252} = \sqrt{36 \cdot 7}$.

Find the power. Write the answer in rectangular form. (3 points)

Problem: $(-2 + 3i)^4$

3. $\tan \theta = -\frac{3}{2}$ is not a unit-circle value, so De Moivre can't give exact values here. Square, then square again.

$$(-2 + 3i)^2 = (-2)^2 + 2(-2)(3i) + (3i)^2 = 4 - 12i + 9i^2 = 4 - 12i - 9 = -5 - 12i$$

$$(-2 + 3i)^4 = (-5 - 12i)^2 = (-5)^2 + 2(-5)(-12i) + (-12i)^2 = 25 + 120i + 144i^2 = 25 + 120i - 144$$

$$= \boxed{-119 + 120i}$$

Trap: $(a + bi)^2 = a^2 - b^2 + 2abi$. Never square each part separately; $(-5)(-12) \cdot 2 = +120$.

Find the fifth roots of the number. Leave answers in trig form with θ in radians. Graph each root as a vector in the complex plane. (3 points)

Problem: $16 - 16i\sqrt{3}$

4. Convert, then roots. Radians throughout.

$$r = \sqrt{16^2 + (-16\sqrt{3})^2} = \sqrt{256 + 768} = \sqrt{1024} = 32$$

$$\tan \theta = \frac{-16\sqrt{3}}{16} = -\sqrt{3} \Rightarrow \text{ref } \frac{\pi}{3}, \text{ QIV} \Rightarrow \frac{5\pi}{3}$$

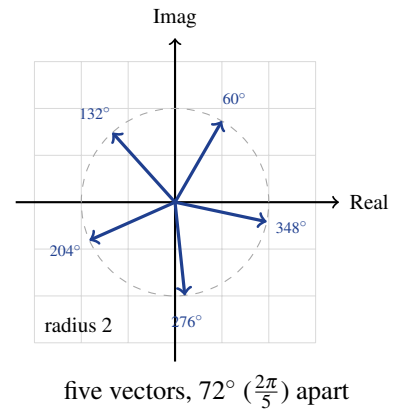
$$\sqrt[5]{32} = 2 \quad \text{first: } \frac{5\pi/3}{5} = \frac{\pi}{3} = \frac{5\pi}{15}$$

$$\text{spacing: } \frac{2\pi}{5} = \frac{6\pi}{15}$$

$$2 \text{cis } \frac{\pi}{3}, 2 \text{cis } \frac{11\pi}{15}, 2 \text{cis } \frac{17\pi}{15}, 2 \text{cis } \frac{23\pi}{15}, 2 \text{cis } \frac{29\pi}{15}$$

Common denominator 15 makes the $+\frac{6\pi}{15}$ steps easy. Five roots, last one $\frac{29\pi}{15} < 2\pi$.

Trap: spacing is $\frac{2\pi}{n}$, not $\frac{2\pi}{3}$ or $\frac{\pi}{5}$.



One of the fourth roots of a complex number w is $-\sqrt{2} + i\sqrt{2}$. (3 points)

Problem: a) Determine the rectangular form of its other three fourth roots.

b) Find w in rectangular form.

5. All n th roots have the same r and sit $\frac{360^\circ}{n}$ apart. Find the given root's trig form, then step around.

$$-\sqrt{2} + i\sqrt{2}: r = \sqrt{2+2} = 2 \quad \tan \theta = -1, \text{ QII} \Rightarrow 135^\circ \quad \text{spacing } \frac{360^\circ}{4} = 90^\circ$$

$$\text{a) } 2 \text{cis } 225^\circ = 2 \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right) = -\sqrt{2} - i\sqrt{2}$$

$$2 \text{cis } 315^\circ = 2 \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right) = \sqrt{2} - i\sqrt{2}$$

$$2 \text{cis } 405^\circ = 2 \text{cis } 45^\circ = 2 \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right) = \sqrt{2} + i\sqrt{2}$$

$$\text{b) } w = (2 \text{cis } 135^\circ)^4 = 2^4 \text{cis } 540^\circ = 16 \text{cis } 180^\circ = 16(-1 + 0i) = \boxed{-16}$$

$$\text{a) } -\sqrt{2} - i\sqrt{2}, \sqrt{2} - i\sqrt{2}, \sqrt{2} + i\sqrt{2} \quad \text{b) } w = -16$$

Trap: fourth roots are 90° apart, not 120° (that is cube roots). To get w , raise the root to the 4th with De Moivre.

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Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[2\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right] \left[\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2. $\frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

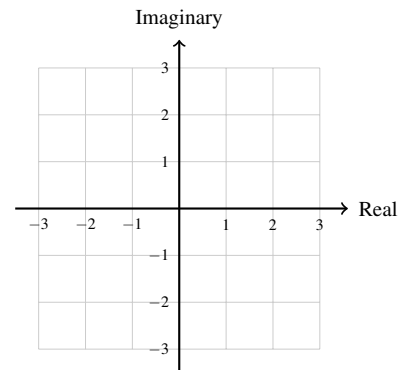
3. $\left[2 \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^8$

Solve. Write answers in rectangular form. (3 points)

4. $x^4 + 81 = 0$

Find the cube roots of the number. Leave answers in trig form. Graph each cube root as a vector in the complex plane. (3 points)

5. $2 - 2i\sqrt{3}$



ANSWER KEY – Non-Calculator

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Every line shown. Point splits are an estimate of partial credit (halves are possible).

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[2\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right] \left[\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right]$

1. Multiply the r 's, add the θ 's. The sum is $\frac{11\pi}{12} = 165^\circ$, which is *not* on the unit circle, so use the sum formula.

$$\begin{aligned} r &= (2\sqrt{2})(\sqrt{2}) = 4 & \theta &= \frac{2\pi}{3} + \frac{\pi}{4} = \frac{8\pi}{12} + \frac{3\pi}{12} = \frac{11\pi}{12} \\ \cos \frac{11\pi}{12} &= \cos \frac{2\pi}{3} \cos \frac{\pi}{4} - \sin \frac{2\pi}{3} \sin \frac{\pi}{4} = \left(-\frac{1}{2}\right) \frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{-\sqrt{2}-\sqrt{6}}{4} \\ \sin \frac{11\pi}{12} &= \sin \frac{2\pi}{3} \cos \frac{\pi}{4} + \cos \frac{2\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} + \left(-\frac{1}{2}\right) \frac{\sqrt{2}}{2} = \frac{\sqrt{6}-\sqrt{2}}{4} \\ 4 \left(\frac{-\sqrt{2}-\sqrt{6}}{4} + \frac{\sqrt{6}-\sqrt{2}}{4} i \right) &= \boxed{-\sqrt{2} - \sqrt{6} + (\sqrt{6} - \sqrt{2})i} \end{aligned}$$

Trap: $\cos(A+B) = \cos A \cos B - \sin A \sin B$ (minus), $\sin(A+B) = \sin A \cos B + \cos A \sin B$ (plus). QII check: $\cos$ negative, $\sin$ positive.

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

Problem: $\frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$

2. Convert top and bottom, divide the r 's, subtract the θ 's.

$$\begin{aligned} \text{Top: } r &= \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4+12} = 4 & \tan \theta &= \frac{-2\sqrt{3}}{-2} = \sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIII} \Rightarrow 240^\circ \\ \text{Bottom: } r &= \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{2+6} = \sqrt{8} = 2\sqrt{2} & \tan \theta &= \frac{-\sqrt{6}}{\sqrt{2}} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ \end{aligned}$$

$$\begin{aligned} \frac{4 \text{cis } 240^\circ}{2\sqrt{2} \text{cis } 300^\circ} &= \frac{2}{\sqrt{2}} \text{cis}(240^\circ - 300^\circ) = \sqrt{2} \text{cis}(-60^\circ) = \sqrt{2} \text{cis } 300^\circ \\ &= \sqrt{2} \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = \boxed{\frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2}i} \end{aligned}$$

Trap: $\frac{2}{\sqrt{2}} = \frac{2\sqrt{2}}{2} = \sqrt{2}$. $\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$. Both parts of the top are negative: QIII, 240° (not 60°).

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $\left[2 \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^8$

3. De Moivre: raise r to the 8th, multiply the angle by 8, then reduce by multiples of 2π .

$$\begin{aligned} 2^8 &= 256 & 8 \cdot \frac{7\pi}{12} &= \frac{56\pi}{12} = \frac{14\pi}{3} \\ \frac{14\pi}{3} - 4\pi &= \frac{14\pi}{3} - \frac{12\pi}{3} = \frac{2\pi}{3} \\ 256 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) &= 256 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = \boxed{-128 + 128\sqrt{3}i} \end{aligned}$$

Trap: $\frac{7\pi}{12}$ is not a unit-circle angle, but $8 \cdot \frac{7\pi}{12}$ reduces to one. Don't reach for the sum formula before multiplying.

Solve. Write answers in rectangular form. (3 points)

Problem: $x^4 + 81 = 0$

4. Isolate x^4 , write the constant in trig form, take fourth roots, convert each.

$$x^4 = -81 = -81 + 0i \Rightarrow r = 81, \theta = 180^\circ \quad x = (81 \text{ cis } 180^\circ)^{1/4}$$

$$\sqrt[4]{81} = 3 \quad \frac{180^\circ}{4} = 45^\circ \quad \text{spacing } 90^\circ \quad x = 3 \text{ cis } 45^\circ, 3 \text{ cis } 135^\circ, 3 \text{ cis } 225^\circ, 3 \text{ cis } 315^\circ$$

$$3 \text{ cis } 45^\circ = \frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i \quad 3 \text{ cis } 135^\circ = -\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$$

$$3 \text{ cis } 225^\circ = -\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i \quad 3 \text{ cis } 315^\circ = \frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i$$

$$x = \pm \frac{3\sqrt{2}}{2} \pm \frac{3\sqrt{2}}{2}i \quad (\text{all four sign combinations})$$

Trap: $x = \pm 3i$ is wrong: $(3i)^4 = 81i^4 = 81$, not -81 . An even power of a pure imaginary is always positive real, so the roots sit between the axes ($45^\circ, 135^\circ, \dots$).

Find the cube roots of the number. Leave answers in trig form. Graph each cube root as a vector in the complex plane. (3 points)

Problem: $2 - 2i\sqrt{3}$

5. Convert, then cube roots. $\sqrt[3]{4}$ does not simplify, so leave it.

$$r = \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

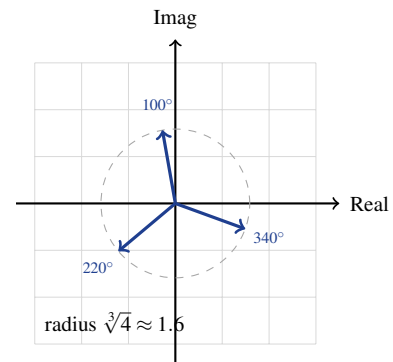
$$\tan \theta = \frac{-2\sqrt{3}}{2} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ$$

$$\sqrt[3]{4} \quad \text{first: } \frac{300^\circ}{3} = 100^\circ \quad \text{spacing } 120^\circ$$

$$\sqrt[3]{4} \text{ cis } 100^\circ, \sqrt[3]{4} \text{ cis } 220^\circ, \sqrt[3]{4} \text{ cis } 340^\circ$$

The graph: three equal vectors of length $\sqrt[3]{4}$ (between 1 and 2), 120° apart (the “peace sign”).

Trap: divide the angle, not the radius: $\sqrt[3]{4}$, not $\frac{4}{3}$. All angles stay positive and under 360° .



three equal vectors, 120° apart